

Shaping What Is Remembered: Initial Study on Object Saliency and Selective Memory Recall Trade-Offs in VR

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Abstract

Object saliency can capture attention and bias what is encoded and later retrieved, yet its effects on memory in immersive virtual reality remain underexplored, particularly when multiple saliency cues appear together in the same scene. In this work, we examine saliency-driven memory bias based on bottom-up attentional capture, rather than goal-directed top-down selection. We built a controlled virtual reality (VR) conference room that contains baseline objects, colour-salient objects, and motion-salient objects within a single mixed environment. Eighteen participants completed a within-subjects learning–interference–recall procedure. Memory was assessed using a three-minute typed free recall task and a twenty-trial old-new recognition test in VR that included category-matched distractors. Initial results show that saliency improved recall and recognition, but also increased false recognition for similar distractors, suggesting a trade-off between stronger encoding and higher susceptibility to memory distortion. These findings inform the design of attention-guiding cues in VR for training, rehabilitation, and immersive interface design.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Empirical studies in HCI*; User studies.

Keywords

Virtual Reality, Object Saliency, Selective Memory Recall, Human Computer Interaction (HCI)

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1 Introduction

In visually rich environments, not all objects are encoded or retrieved equally. Object-level saliency can capture attention and

bias memory outcomes, making some items more likely to be remembered than others. This function is relevant to everyday activities such as hazard detection [1, 4, 13, 35], medication management, and navigation, and is often impaired in conditions including Alzheimer’s disease and traumatic brain injury [2, 23]. Despite extensive memory research, how bottom-up attentional capture contributes to later memory benefits and memory errors remains an open question [18].

Object saliency can capture attention and bias encoding toward salient items [7, 20, 32]. Virtual reality (VR) provides an ecologically valid yet controllable environment for manipulating saliency cues and observing memory behaviour during natural exploration [17, 34]. In this study, we focus on saliency-driven memory bias in VR: how object-level saliency cues (colour vs. motion) shape encoding and retrieval outcomes in a mixed scene. Because participants were instructed to remember as many objects as possible, our design examines bottom-up attentional capture rather than goal-directed top-down selection between task-relevant and task-irrelevant items.

Our contributions are the following: We (1) present a controlled VR paradigm to study saliency-driven memory bias under free exploration; (2) compare baseline, colour salience, and motion salience objects within a learning–interference–recall procedure; and (3) report initial evidence that saliency improves recall and recognition while increasing false recognition to category-matched distractors.

2 Related Work

Memory performance has long been studied in cognitive psychology as a process shaped by attention and encoding, which determines what information is retained and later retrieved. In particular, saliency-related work has shown that distinctive perceptual features can produce bottom-up attentional capture and bias encoding toward salient items, which can in turn influence later recall and recognition [2, 7, 20, 23, 32]. Prior behavioural and neurocognitive studies suggest that salient features are often preferentially encoded and recalled, but the same mechanisms may also contribute to memory distortions when retrieval relies on familiarity or incomplete feature binding.

Virtual reality (VR) provides an ecologically valid yet controllable environment for studying attention and memory in more naturalistic scenes. Recent work has explored emotional and autobiographical memory in VR, including how immersive contexts and multimodal feedback shape memory processes [3, 9–12, 21, 24]. Collectively, these studies demonstrate the value of VR for designing controllable yet immersive memory paradigms, but they primarily focus on affective or autobiographical memory rather



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than saliency-driven object memory bias in mixed scenes [6]. Compared with static laboratory tasks, VR offers ecological validity and repeatability, providing unique opportunities to study how saliency cues shape both memory benefits (improved recall/recognition) and memory costs (false recognition) during free exploration [5, 16, 18, 27, 37]. This is the gap that our work addresses.

3 Initial Study

3.1 Experimental Design

The goal of the initial study is to investigate how different object saliency cues (colour and motion), when co-present in an immersive VR environment, produce saliency-driven memory bias in recall and recognition, including potential trade-offs such as false memories. To evaluate this, all participants are required to explore the same VR conference room scene containing baseline, colour-salient, and motion-salient objects. Saliency type (baseline vs. colour vs. motion) served as the primary independent variable, as shown in figure 1 and figure 3. Because each participant was exposed to all three saliency categories, the study follows a within-subject (repeated-measures) design. Primary dependent measures included (1) free-recall performance and (2) recognition performance (hit rate and false-alarm rate). Additionally, we recorded attention-related behavioural measures during exploration.

3.2 Procedure

The procedure consisted of three phases, as shown in figure 2:

Learning phase. Participants freely explored the meeting-room scene for 5 minutes and were instructed to remember as many objects as possible. Attention allocation was self-directed while participants walked naturally and rotated their heads to inspect objects.

Interference phase. After learning, participants completed a 5-minute mental arithmetic task (addition/subtraction) without a calculator. This phase was used to reduce immediate rehearsal effects and better reflect delayed retrieval.

Recall phase. This phase is divided into Free Recall and Recognition. During Free Recall, participants typed the names of the objects they remembered on a computer within 3 minutes. Synonyms were accepted and scored as correct matches. During Recognition, participants then returned to the VR scene and completed a 20-trial old/new recognition task, comprising 10 targets (previously seen) and 10 same-category distractors (unseen; e.g., apple → pear/orange; computer → keyboard/monitor). Items were presented as 3D objects in VR. The task was untimed, and we recorded accuracy only (no reaction times).

3.3 Participants and Setup

We recruited 18 participants (18–25 years; 10 female). Participants provided informed consent and completed the study under a standardised setup using a Meta Quest 3¹ headset and a Unity-based² conference room scene. Participants explored the room by physical walking and moving their heads to look around, without any controller-based locomotion to preserve immersion and realism.

¹<https://www.meta.com/quest/quest-3/>

²<https://unity.com/>



Figure 1: Scene of a participant exploring the same virtual room containing baseline, colour-salient (emissive), and motion-salient (subtle motion) objects.

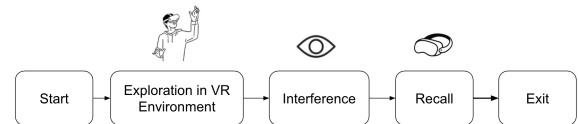


Figure 2: Procedure of the initial study

3.4 Behavioural and Attention Measures

We recorded attention-related behavioural data during VR exploration. Because Meta Quest 3 does not provide native eye tracking, we used head-gaze raycasting as a proxy for attention. Specifically, we inferred attended objects from raycast, object hits derived from head orientation and computed metrics such as dwell/fixation time per object. These measures were used to examine whether saliency systematically biases attention allocation and how attention relates to memory outcomes (recognition hits and false recognitions).

3.5 Statistical Analysis

We use a repeated measures ANOVA with saliency type (baseline/colour/motion) as a within-subject factor to test differences in memory performance across object types. Primary outcomes include free-recall performance and recognition performance (hit rate and false alarm rate). If significant effects are observed, we will conduct post hoc pairwise comparisons with appropriate multiple-comparison correction and report effect sizes. Where needed, we will assess sphericity and apply corrections.

4 Initial Results

The analysis of memory performance across the three saliency conditions (Baseline, Colour Saliency, and Motion Saliency) revealed a consistent monotonic pattern (Table 1). Free recall accuracy (RA) increased from Baseline to Colour and further to Motion, with Baseline showing the lowest recall (RA = 0.36, SD = 0.12), Colour

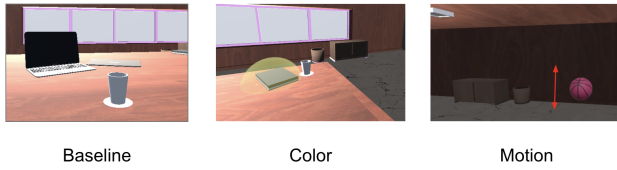


Figure 3: Three saliency conditions (left to right): Baseline (no effect), Color (fluorescent halo, 1.5× object extent), and Motion (periodic vertical bouncing).

Table 1: Memory and attention outcomes by saliency condition (mean (SD)). RA = recall accuracy (proportion). HR = recognition hit rate. FRR = false recognition rate (false alarm rate). DT = mean dwell time per object (s).

Measure	Baseline	Color	Motion
RA (proportion)	0.36 (0.12)	0.54 (0.12)	0.74 (0.10)
HR	0.68 (0.11)	0.81 (0.08)	0.90 (0.06)
FRR	0.18 (0.09)	0.22 (0.10)	0.30 (0.11)
d'	1.40 (0.45)	1.61 (0.42)	1.86 (0.40)
DT (s)	3.5	4.12	4.9

yielding intermediate performance (RA = 0.54, SD = 0.12), and Motion achieving the highest recall (RA = 0.74, SD = 0.10). A similar trend was observed in recognition performance: hit rate (HR) rose from Baseline (0.68, SD = 0.11) to Color (0.81, SD = 0.08) and Motion (0.90, SD = 0.06), and sensitivity (d') also increased accordingly (Baseline: 1.40, SD = 0.45; Color: 1.61, SD = 0.42; Motion: 1.86, SD = 0.40).

Importantly, stronger saliency cues were accompanied by an increase in false recognition. The false recognition rate (FRR) increased from Baseline (0.18, SD = 0.09) to Colour (0.22, SD = 0.10), and was highest in Motion (0.30, SD = 0.11). This pattern suggests a trade-off: while motion-based saliency produced the strongest memory benefits (higher RA, HR, and d'), it also increased the tendency to misidentify category-matched distractors as previously seen. Attention-related behaviour showed a matching trend, with mean dwell time (DT) increasing from Baseline (3.5 s) to Colour (4.12 s) and Motion (4.9 s), indicating that more salient objects attracted more sustained attention during encoding.

Subjective feedback may help contextualise these patterns. For example, participants may report that motion cues were highly attention-grabbing and made salient objects easier to remember, but sometimes led to confusion when distractors were visually or semantically similar. Colour cues may be described as helpful for quick identification and stable tracking of objects, whereas the baseline condition may encourage category-based or associative encoding strategies.

5 Discussion

This study examines how different saliency cues influence memory outcomes in VR, with a focus on colour-based and motion-based

saliency. Conceptually, our findings are best interpreted as saliency-driven memory bias based on bottom-up attentional capture, rather than goal-directed top-down selection. Overall, our results suggest a clear benefit-cost trade-off in memory performance.

5.1 Saliency and memory encoding

Our results showed that motion saliency improved recall accuracy relative to the neutral condition, whereas colour saliency showed a weaker benefit [19, 22, 26]. This pattern is consistent with prior work showing that salient stimuli are more likely to capture attention and be encoded effectively during learning. Motion saliency may be especially effective because it remains perceptually prominent during exploration, supporting rapid processing and stronger encoding. In this sense, motion-based cues can increase perceptual distinctiveness and attract bottom-up attention, thereby improving later retrieval [23].

5.2 Saliency and false recognition

Another key contribution of this study is the observation of saliency-induced false recognition. Although saliency improved recall and recognition for target items, it also increased false alarms for category-matched distractors. This suggests that stronger saliency cues can improve memory accessibility while also increasing retrieval confusion when alternatives are perceptually or semantically similar.

One possible explanation is that saliency strengthens familiarity-based judgments, making similar lures feel previously seen [8, 28]. Saliency may also narrow attention during encoding, leading to weaker encoding of diagnostic details needed to distinguish targets from look-alikes [14, 30, 31]. In addition, the apparent advantage for salient objects may reflect both enhanced encoding of salient items and relatively weaker encoding of non-salient co-present items. Our current design does not directly separate these mechanisms. Together, these findings suggest that saliency is a double-edged design tool in VR.

5.3 Applications, limitations, and future directions

These findings have practical implications for VR design. In education, training, and rehabilitation, saliency cues may help direct attention to important content and improve memorability. However, the same cues may also increase false recognition when similar alternatives are present. Designers should therefore use saliency selectively and calibrate it to balance memory benefits with memory reliability [15, 25, 29, 33, 36].

This study also has several limitations. First, the sample size was modest ($N = 18$), which limits statistical power and generalisability. Second, attention was approximated using head-gaze raycasting rather than eye tracking, which provides a practical proxy but cannot capture fine-grained eye movement behaviour. Third, the study used a single conference-room scene and a short observation window, so the temporal dynamics of saliency (e.g., habituation) and the scalability of the findings remain unclear. Future work should use larger samples, eye-tracking-enabled VR hardware, and multiple scenes to test the robustness of the observed trade-off.

6 Conclusion and Future Works

This work presents a VR system for studying how object saliency shapes memory in an immersive, freely explored scene. Rather than testing goal-directed selective recall, the present design captures saliency-driven memory bias under bottom-up attentional capture. Stronger saliency cues improved recall and recognition, but also increased false recognition for category-matched distractors, revealing a clear benefit-cost trade-off. Overall, saliency can strengthen memory accessibility while biasing retrieval toward familiarity when similar alternatives are present, suggesting that saliency in educational or rehabilitative VR should be calibrated to support remembering without amplifying false memories.

Future work will (1) systematically vary distractor similarity to pinpoint when saliency begins to inflate false recognition, (2) add confidence ratings and reaction times to better separate familiarity driven errors from deliberate decisions, and (3) link false alarms to gaze and exploration logs (e.g., fixation duration and revisits) to explain how attention during encoding predicts later memory outcomes.

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Generative AI tools were used at the paragraph level to improve readability and style of text originally envisioned and written by the authors, and to adjust figures (e.g., improving clarity, layout, or labeling) based on author-created originals. The AI assistance was limited to copy-editing and visual refinement; all conceptual content, study design, analysis, and interpretation were created by the authors. All AI-assisted text and image adjustments were carefully reviewed, revised, and approved by the human authors, who remain fully responsible for the final manuscript. No confidential or proprietary data were provided to the AI system.

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